高速ソフトウェアルータ(Kamuee)による GTP-U/SRv6のStateless変換の遅延評価

MPLS JAPAN 2020

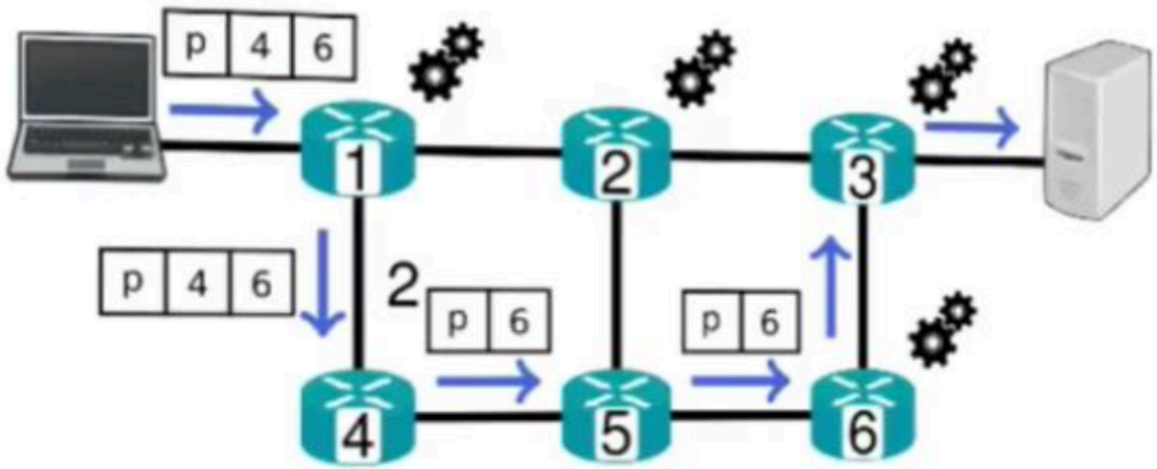
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Nov.9th.2020

Toyota Motor Corp.[1)], Intel K.K. [2)], NTT Communications Corporation[3)]
Arrcus, Inc.[4)], Fujitsu Laboratories Limited[5)], SoftBank Corp.[4)]

Introduction

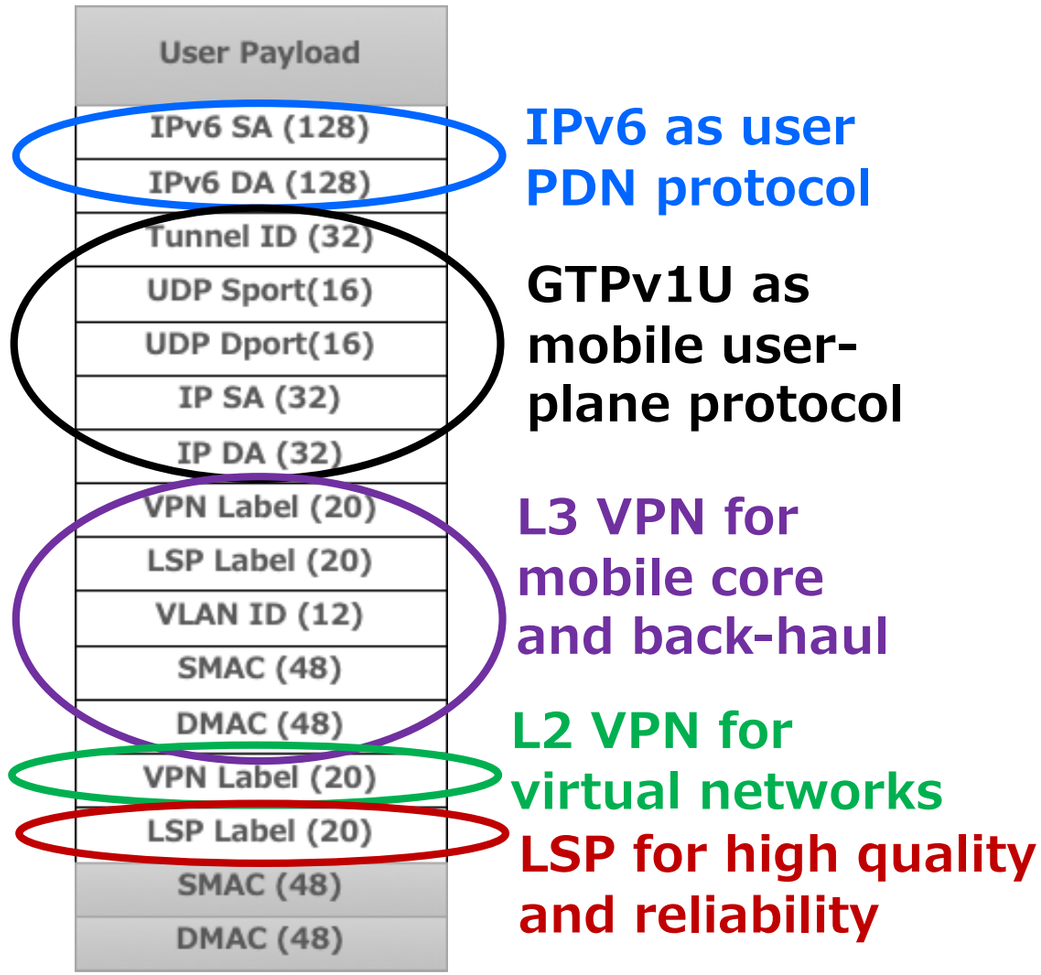
- Segment routing IPv6 (SRv6)
  - Encode the path in each packet



SRv6 based on source routing has many advantages: stateless traffic steering, state reduction, network programming and so on

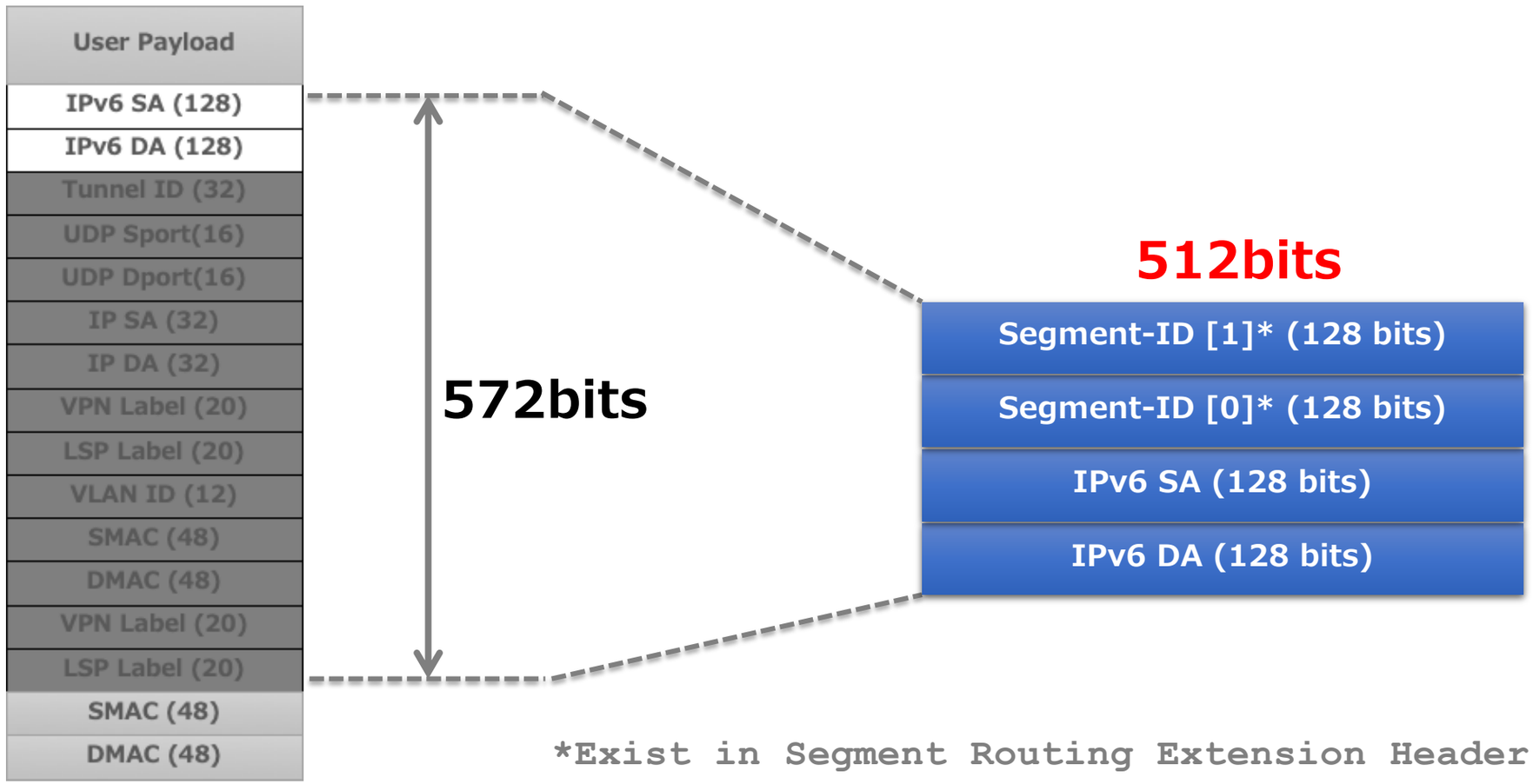
Why the translation is required? – (1)

- The data plane of mobile network has multiple stacking headers
  - Stacking multiple small IDs to fulfill requirements of reliability, VPNs, etc.,
  - **After network failure, the efficient network path would not be selected due to the state of stacking headers**



Why the translation is required? – (2)

- Consolidate all layer headers into one single IPv6 layer

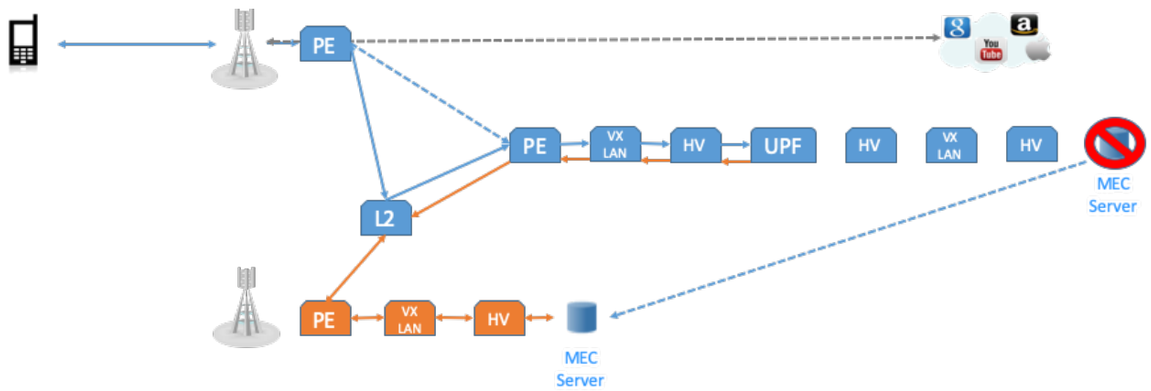


*Exist in Segment Routing Extension Header (SRH)

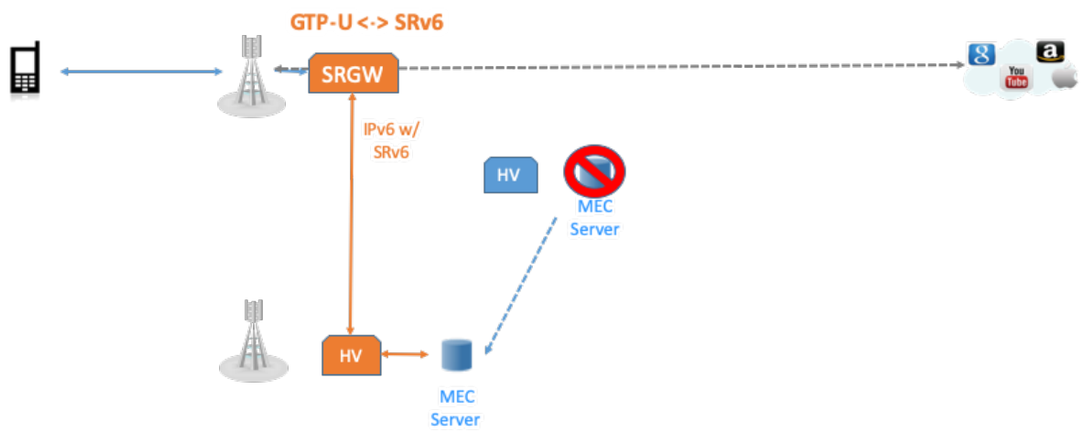
GTP-U/SRv6 translation for mobile network

- Mobile devices are only communicated with UPF (GTP-U) on LTE network
- Using Segment routing gateway (SRGW), the stacking headers can be embedded to SRH and the efficient path can be selected

<Current LTE >

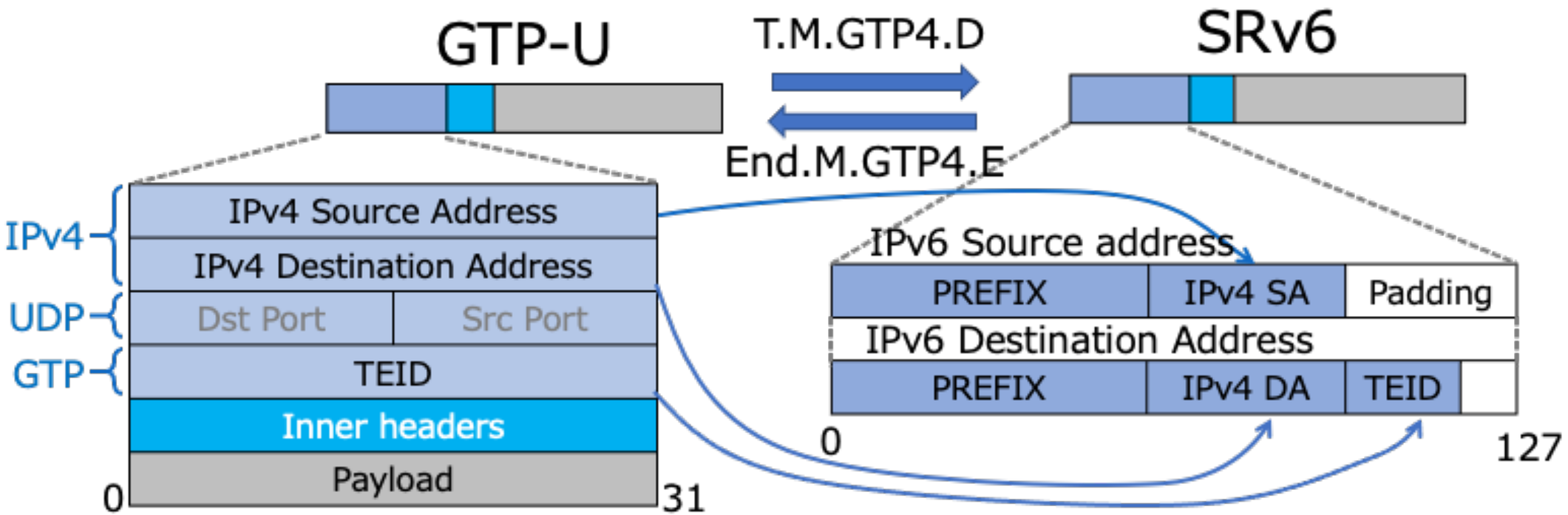


<LTE + SRGW>



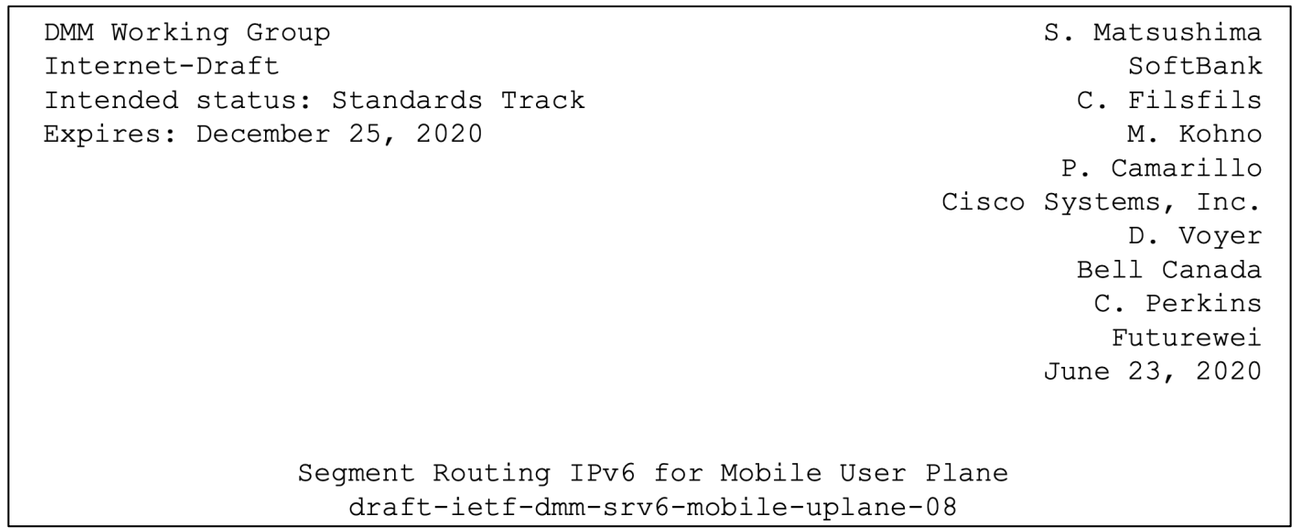
What is GTP-U and SRv6 Stateless Translation?

- All identifiers of a GTP-U tunnel (IPv4 info. and TEID) can be embedded as an argument of a Segment ID (SID)



Current state of translation method

- The method (SRv6 for Mobile UserPlane) has been proposed in IETF
  - https://tools.ietf.org/html/draft-ietf-dmm-srv6-mobile-uplane-08



DMM Working Group
Internet-Draft
Intended status: Standards Track
Expires: December 25, 2020

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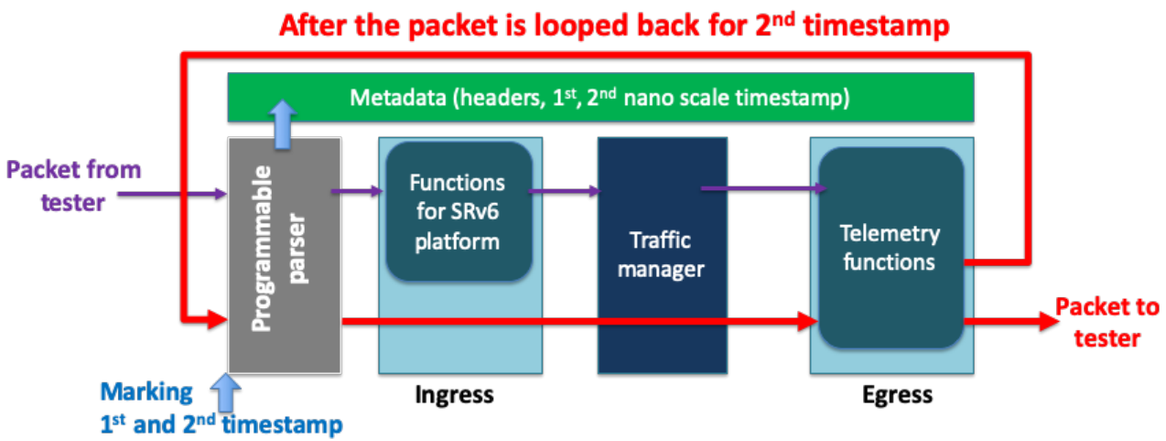
Segment Routing IPv6 for Mobile User Plane
draft-ietf-dmm-srv6-mobile-uplane-08

However, there are no suitable measurement platform and performance evaluation results between GTP-U and SRv6

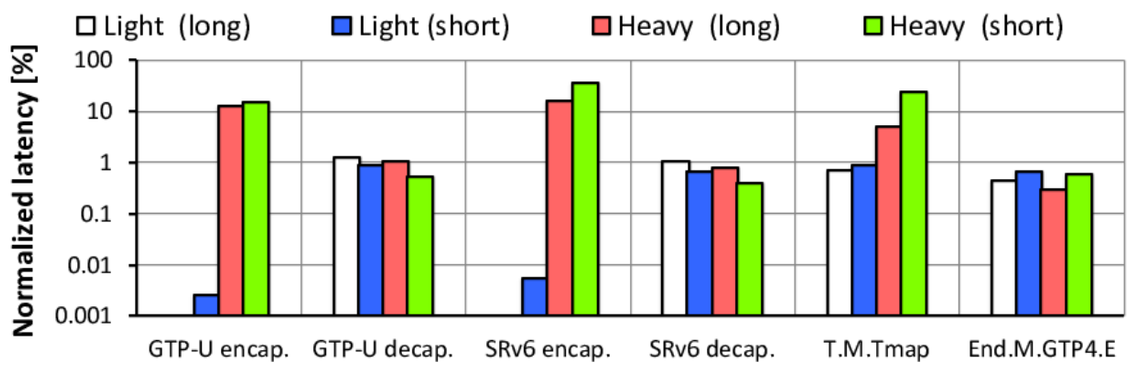
Our previous work[1)]…

- We implemented the translation functions on the H/W P4 switch and measured the latency on the H/W P4 switch

However, the SRv6 Mobile UserPlane with the P4 switch would be hard to deploy Edge/MEC or local 5G network



< SRv6 functions on P4 switch >



< Normalized latency at SR functions >

1) "Performance Evaluation of GTP-U and SRv6 Stateless Translation", 2nd Workshop on Segment Routing and Service Function Chaining (SFC 2019)

Research goal

- Evaluate the quantitative performance of SRv6 Mobile User Plane on DPDK-based software router (Kamuee)

- Observe the latency characteristics on the software router (Kamuee)

How to measure the precise latency on the software router?

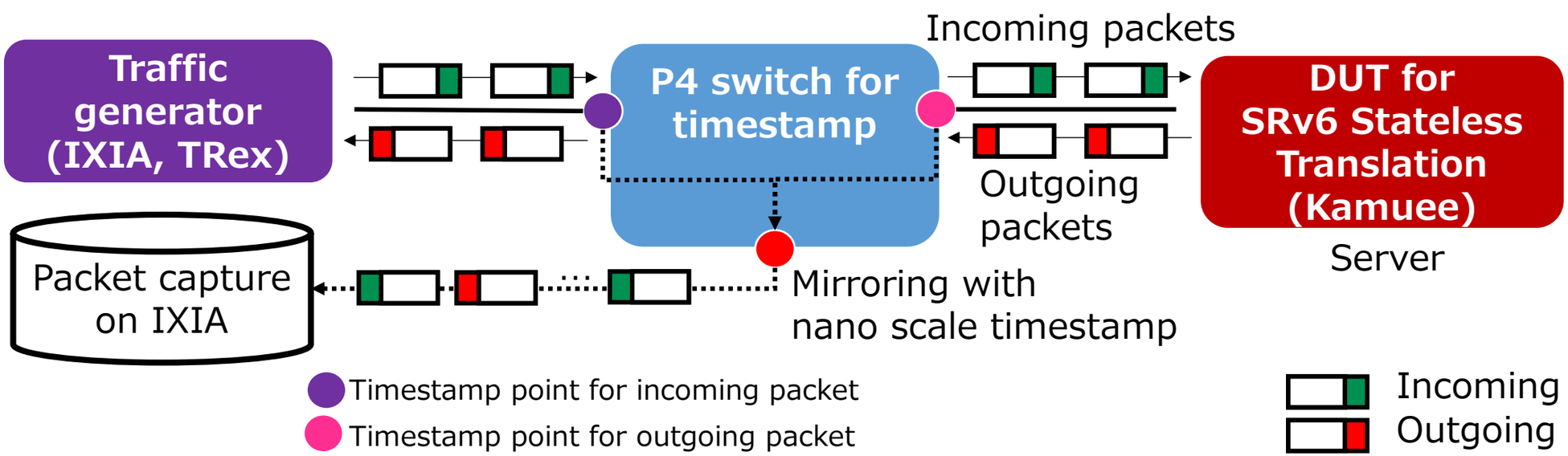
1. How can we preciously measure latency when a receiving packet type is changed from the corresponding packet type sent?
   → **Use nano scale timestamp injection in the P4 switch as telemetry with a pair of unique inner headers**

2. What kind of latency characteristics do we observe on the software router?
   → **Analyze the latency with *rte_eth_tx_burst* and an outgoing packet interval on Kamuee**

Overview

Experiments on software router

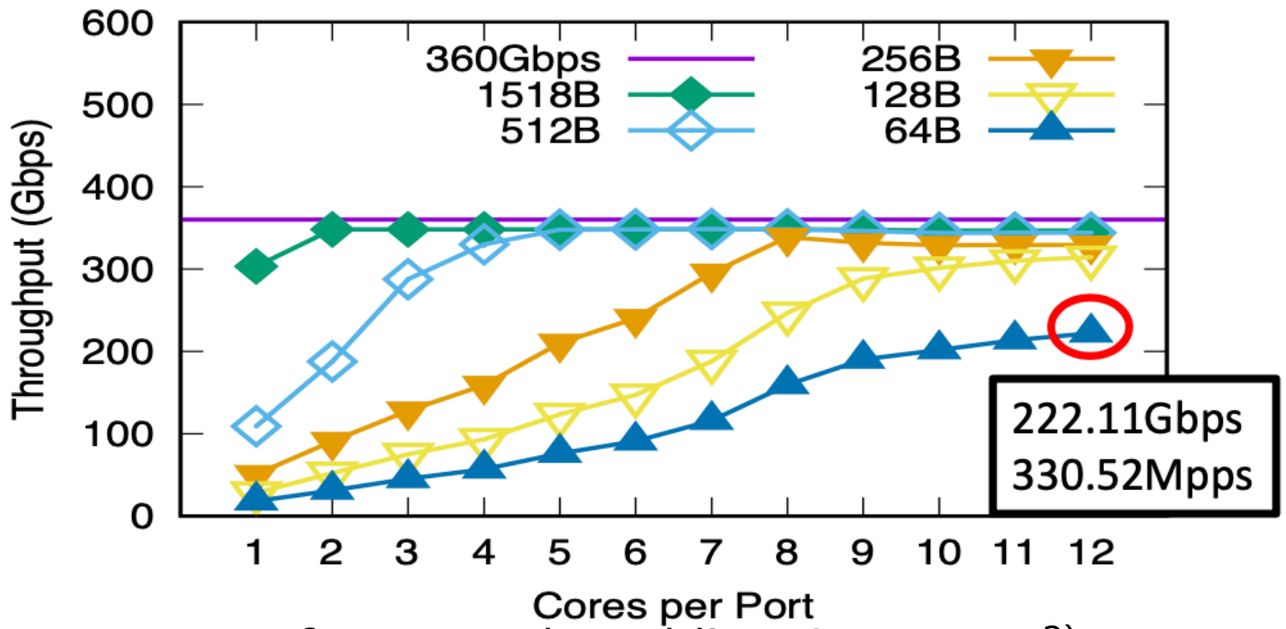
- SRv6 translation functions are used for the latency measurement
  - DUT : DPDK-based software router (Kamuee) is used
  - Traffic generators
    - ***IXIA*** : a hardware-based commercial traffic generator product
    - ***TRex*** : an DPDK based open source traffic generator
  - P4 switch : Inject timestamp as telemetry to source mac address at mirrored packets



Overview of software router

Kamuee – (1)

- A simple DPDK router that evolved from the l3fwd example application of the DPDK
- Kamuee successfully increases its performance as the number of CPU cores increase (input throughput : 360 [Gbps])



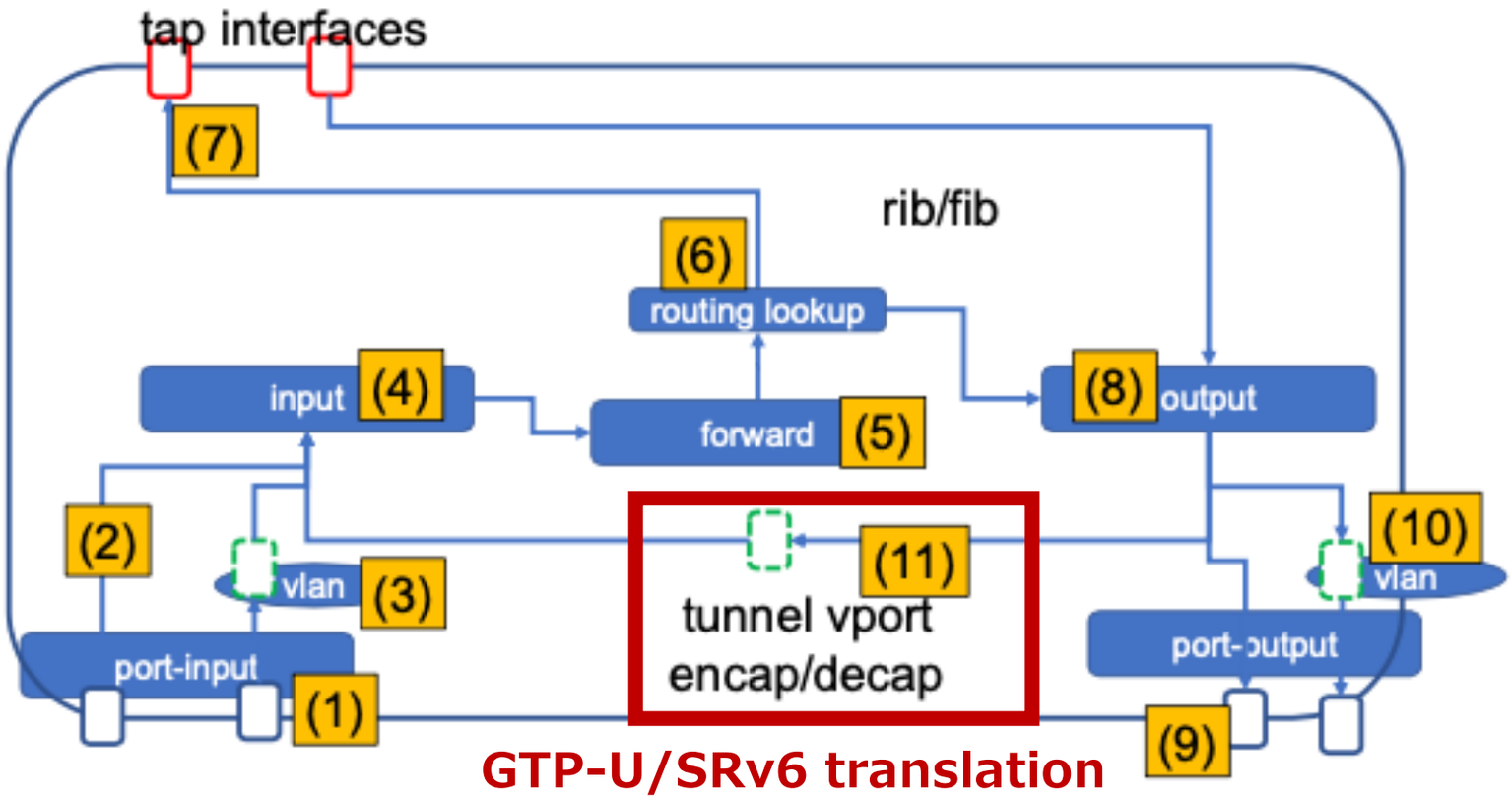
< Performance by adding CPU cores[2] >

2) Kamuee: An IP Packet Forwarding Engine for Multi-Hundred-Gigabit Software-based Networks プロシーディング集 Internet Conference (IC) 2018

Overview of software router

Kamuee – (2)

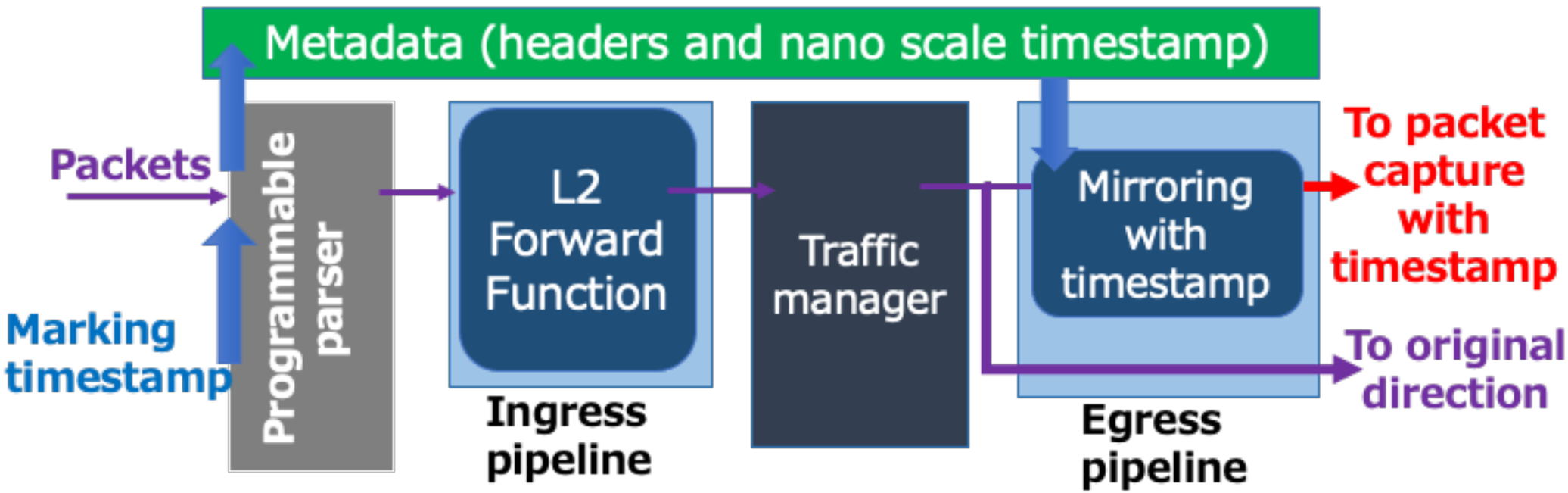
- Kamuee processes on a packet-by-packet basis that rather reflects an input traffic pattern
- GTP-U and SRv6 stateless translation is occurred at tunnel vport (11)



Precise latency measurement

Nano scale timestamp using P4 switch

- During the packet mirroring, the timestamp is stored in the source MAC address field at the outer header
  - This function overwrites and abandons the original source MAC address

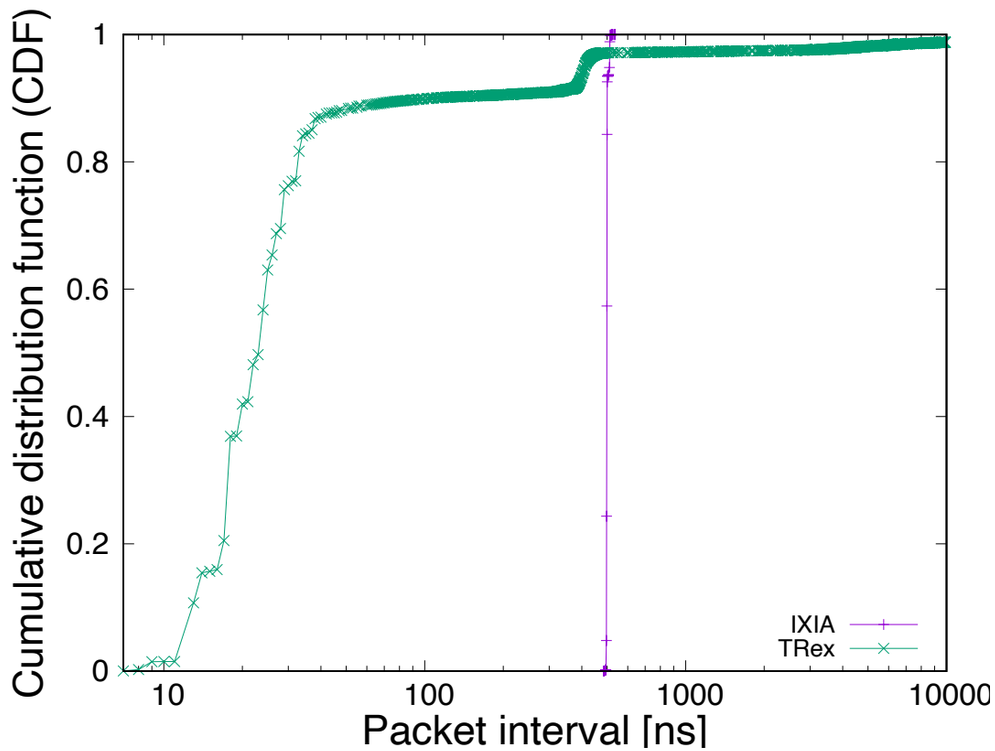


Traffic generators (IXIA and TRex)

CDF of incoming packet interval

- The CDF curves of IXIA and TRex are different, and the curve of TRex is widely fluctuated
  - The traffic generated by TRex is more bursty in constant traffic load (2 [Mpps])

	Configured [ns]	Min [ns]	50%tile [ns]	Mean [ns]	Max [ns]
IXIA	500	484	499	500	537
TRex	500	**7**	24	500	**88,291**



Measurement scenarios

- Focus on the precise latency measurement on Kamuee
- To achieve it, we only used one CPU core for packet processing thread with one NIC port at constant Mpps (2 Mpps)
  - The input load (2 Mpps) is close to maximum throughput (2.5 Mpps)
  - Two traffic gen. (IXIA and TREX) are used
  - Short packet size : ***94 bytes (GTP-U)/98 bytes (SRv6)***
  - Long packet size : ***976 bytes (GTP-U)/980 bytes (SRv6)***

Packet size	Input Load [Mpps]	Number Of trials	Captured packets	GTP-U→SRv6		SRv6→GTP-U	
				RX [Gbps]	TX [Gbps]	RX [Gbps]	TX [Gbps]
Short	2	5	100,000	1.50	1.57	1.57	1.50
Long	2	5	100,000	15.55	15.68	15.68	15.55

Measurement results on Kamuee

Evaluation results

- Statistics of translation latency
  - The latency at all conditions is smaller than 70 μs
  - We cannot observe significant difference between the packet sizes in comparison with the generators
  - The latency with TREX is widely fluctuated
    - Major cause of fluctuated latency is a wide range of input packet interval
    - The interval is affected to the latency measurement

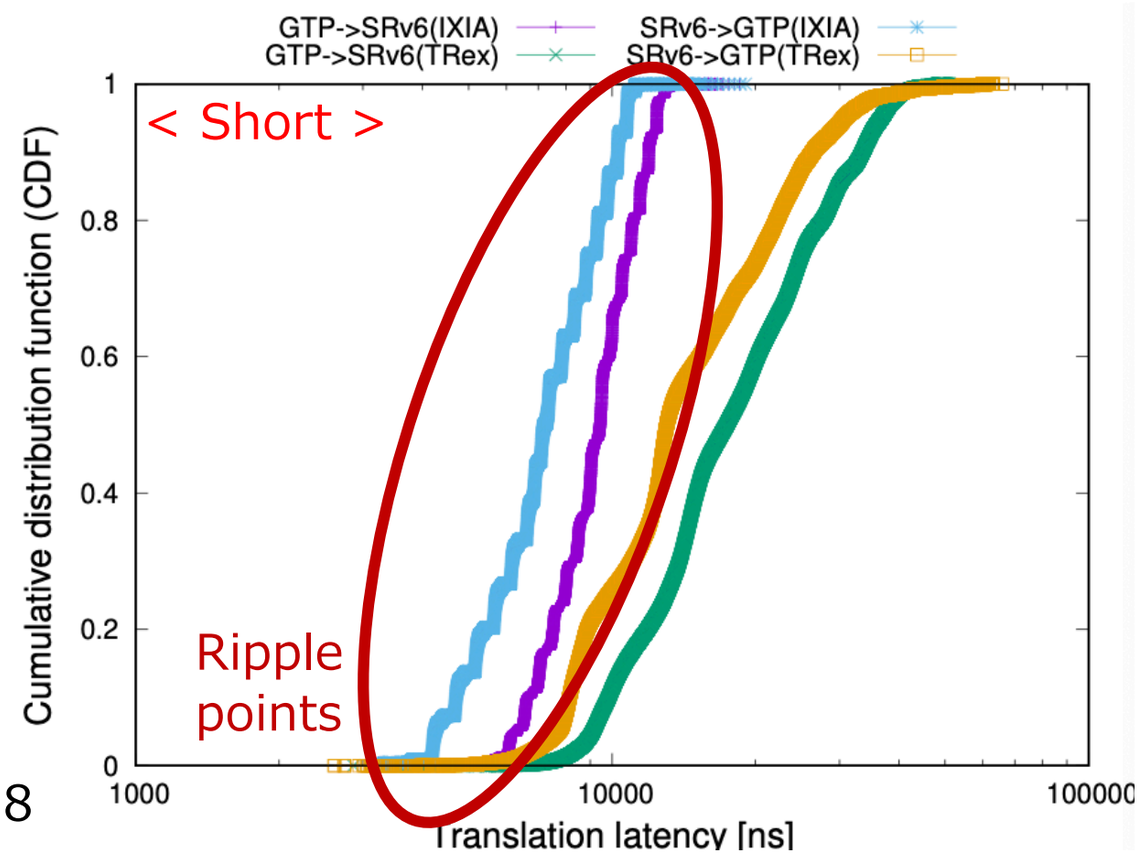
      → **It would be hard to understand the latency measurement results**

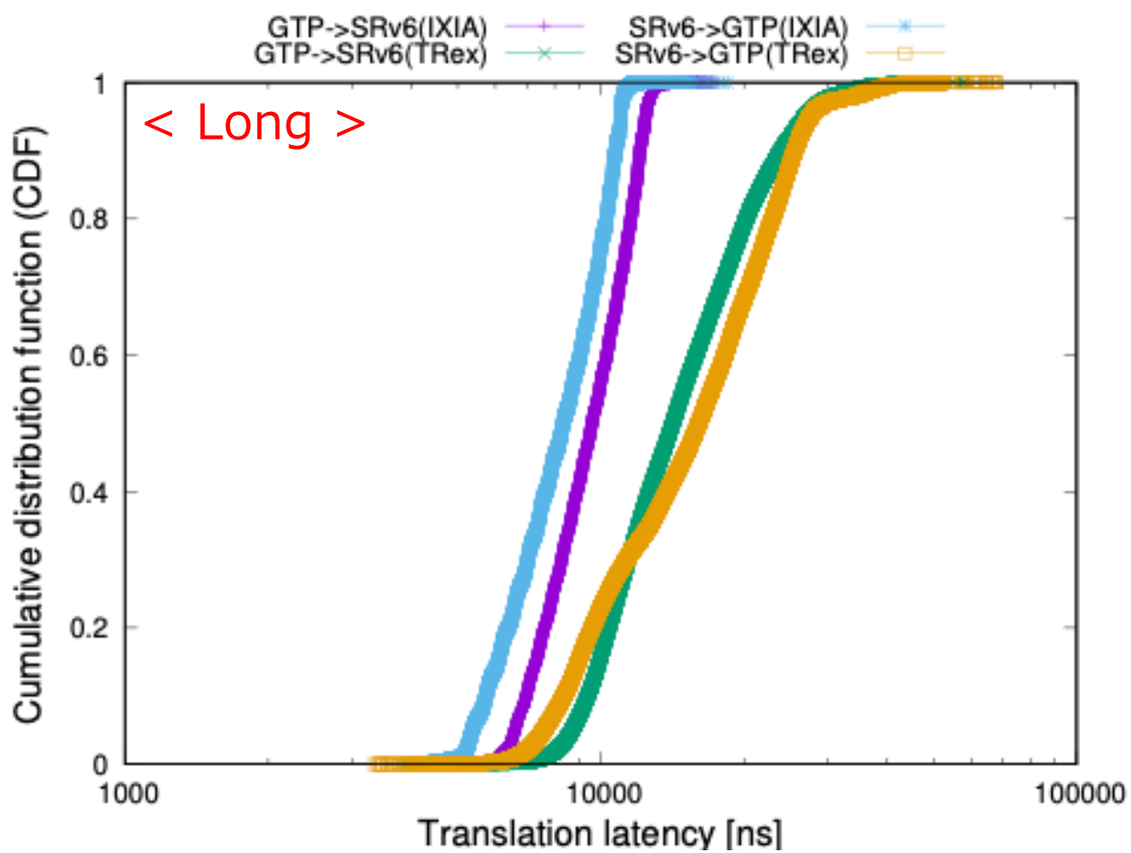
SR function	Traffic Gen.	**Short packet size**				**Long packet size**			
		Min [us]	50%tile [us]	Mean [us]	Max [us]	Min [us]	50%tile [us]	Mean [us]	Max [us]
GTP-U→SRv6	IXIA	4	9	9	19	5	10	10	18
	TRex	3	17	20	51	4	14	16	58
SRv6→GTP-U	IXIA	3	7	7	19	4	8	8	16
	TRex	3	13	16	66	3	16	17	68

Measurement results on Kamuee

CDF of translation latency

- Observe a difference between the generators
  - There are ripple points on the short packet in the CDF curve (IXIA)





Why the ripple points are occurred?

The impact of ***rte_eth_tx_burst***

- What is ***rte_eth_tx_burst***?
  - The packets are sent in a bulk on Kamuee after the translation
- Observe the impact of ***rte_eth_tx_burst*** to the latency measurement
  - IXIA is only used for the precise latency measurement
  - The following tx_burst values are used

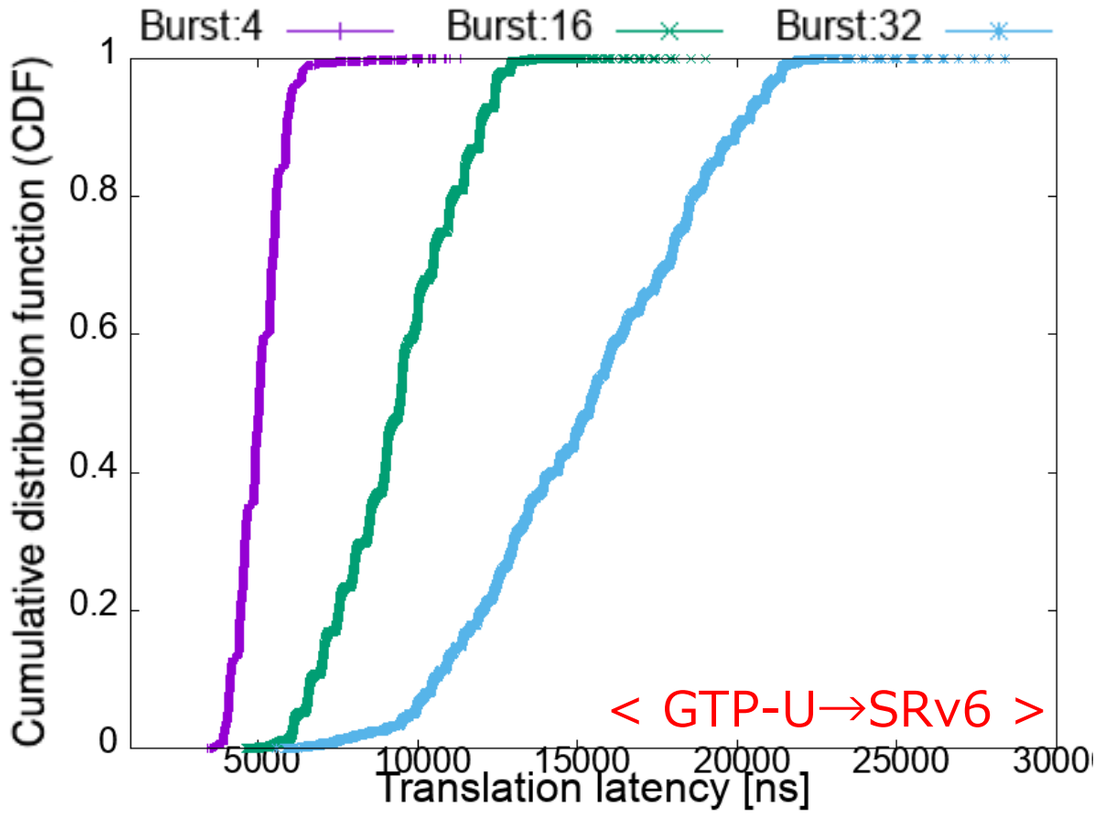
rte_eth_tx_burst	4	16 (default)	32

Why the ripple points are occurred?

CDF of translation latency (short packet)

- Under the input load (2Mpps), the small size of ***tx_burst*** contributes to the short latency
- As the ***tx_burst*** size is changed, the latency is proportionately changed
  - The ***tx_burst*** size is the major cause of ripple points

Input load [2 Mpps]	tx_burst	Min [us]	50%tile [us]	Mean [us]	Max [us]
GTP-U→SRv6	**4**	**3**	**5**	**5**	**11**
	16	4	9	9	19
	32	5	15	15	28
SRv6→GTP-U	**4**	**2**	**3**	**3**	**11**
	16	3	7	7	19
	32	3	12	12	25

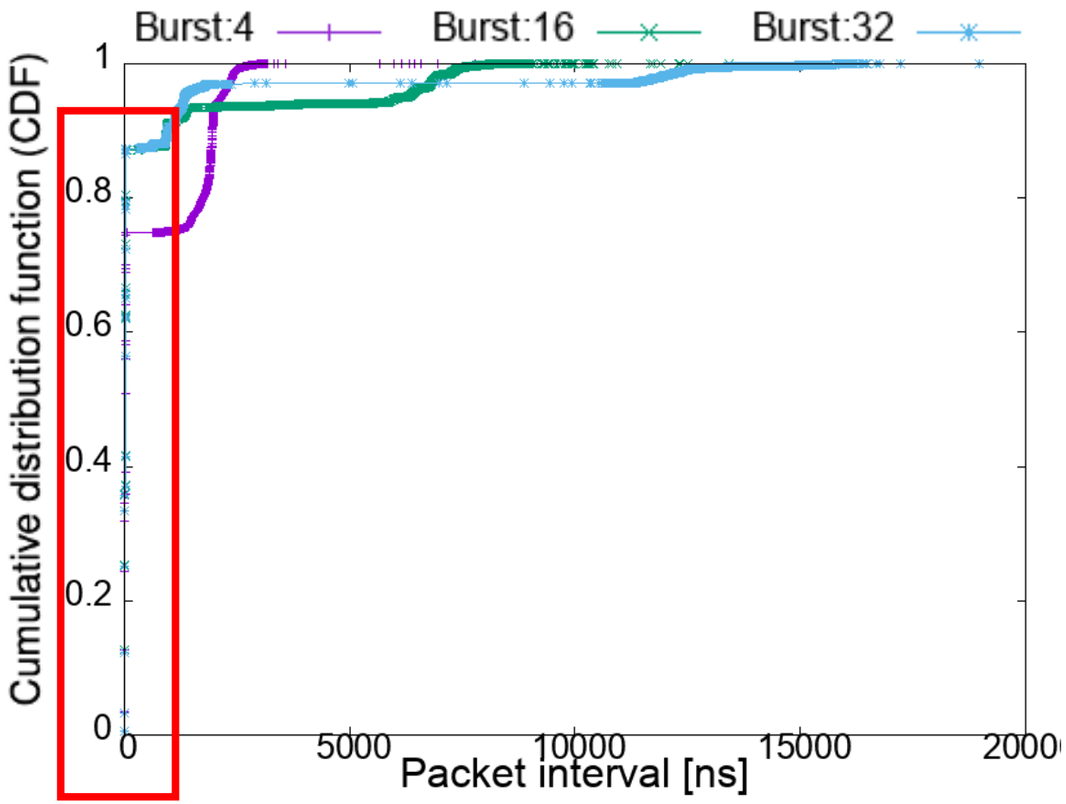


Why the ripple points are occurred?

CDF of outgoing packet interval (short packet)

- Approximately 80% of interval is in the bulk
- The tail interval (i.e. 99%tile) is increased due to the ***tx_burst*** size

Input load [2 Mpps]	tx_burst	Min [ns]	50%tile [ns]	Mean [ns]	Max [ns]
GTP-U→SRv6	4	7	18	500	7,793
	16	7	18	500	13,410
	32	7	18	500	18,979
SRv6→GTP-U	4	7	21	500	8,461
	16	6	16	500	15,351
	32	6	13	500	18,872



Conclusion and future work

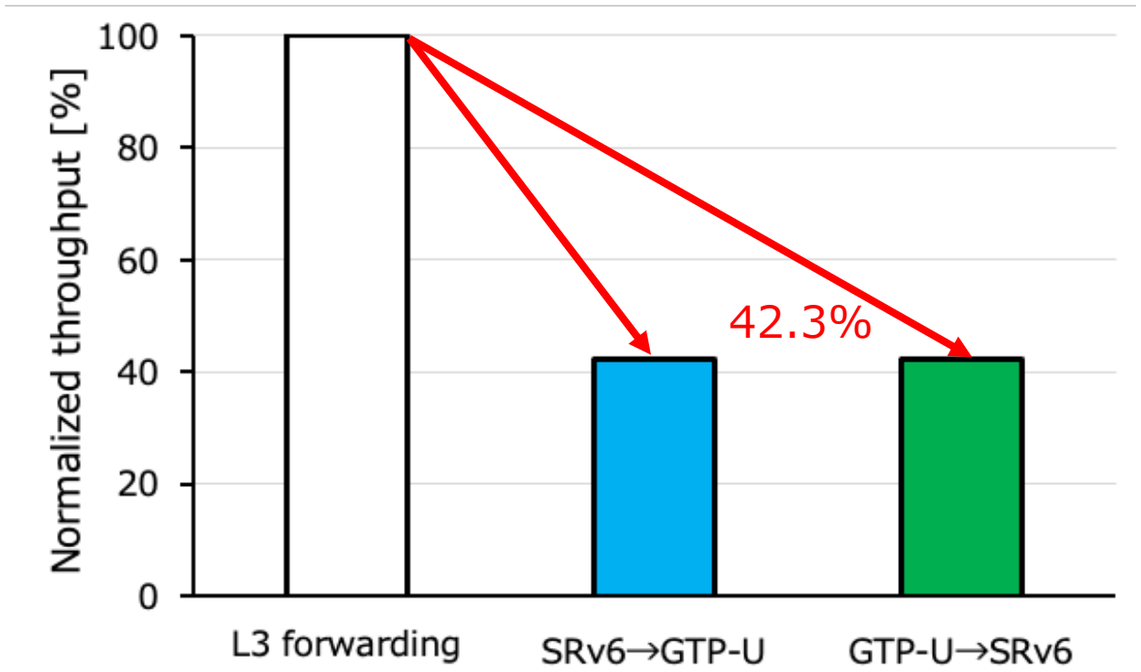
- **Conclusion**
  - We evaluated the latency of translation functions on Kamuee using nano-scale measurement with P4 switch
    - The latency at one CPU core is in the range (roughly 3–70 μs)
    - ***TREX***
      - The latency is widely distributed by the wide range of incoming packet interval
    - ***IXIA***
      - There are ripple points on the short packet
      - The major cause of ripple points is the ***tx_burst*** size
      - When the ***tx_burst*** size is changed, the latency is proportionately changed
- **Future work**
  - Measure the latency on more complicated cases, such as multiple routing entries and multiple CPU cores at maximum performance
  - Evaluate the latency on other software routers

Backup slide

L3 forwarding vs Translation

Comparison of maximum throughput [Mpps]

- Measure the maximum throughput (L3 forwarding [IPv4] vs Translation)
  - There is **NO** packet loss and the single CPU core is used
  - L3 forwarding [IPv4]: 5.9 Mpps
  - GTP-U/SRv6 Translation : 2.5 Mpps



Latency measurement setup (server)

- H/W specification
  - One CPU core is used for latency measurement

CPU	Intel Xeon Gold 6126 (2.6 GHz) [19.25 MB L3 cache]
Memory	384 GB
NIC	Mellanox ConnectX-4
Bandwidth	40 Gbps

- S/W configuration
  - C state/P state are disable
  - CPU frequency is fixed (2.6 GHz)
    - Intel TurboBoost is also disable
  - ***rte_eth_tx_burst*** on Kamuee
    - default : 16

OS	Ubuntu 18.04.2 LTS
Kernel	4.15.0-45
DPDK	18.11.2 (stable)

How to generate unique packets for latency measurement?

- We fixed outer headers and changed unique inner packet headers randomly and sequentially

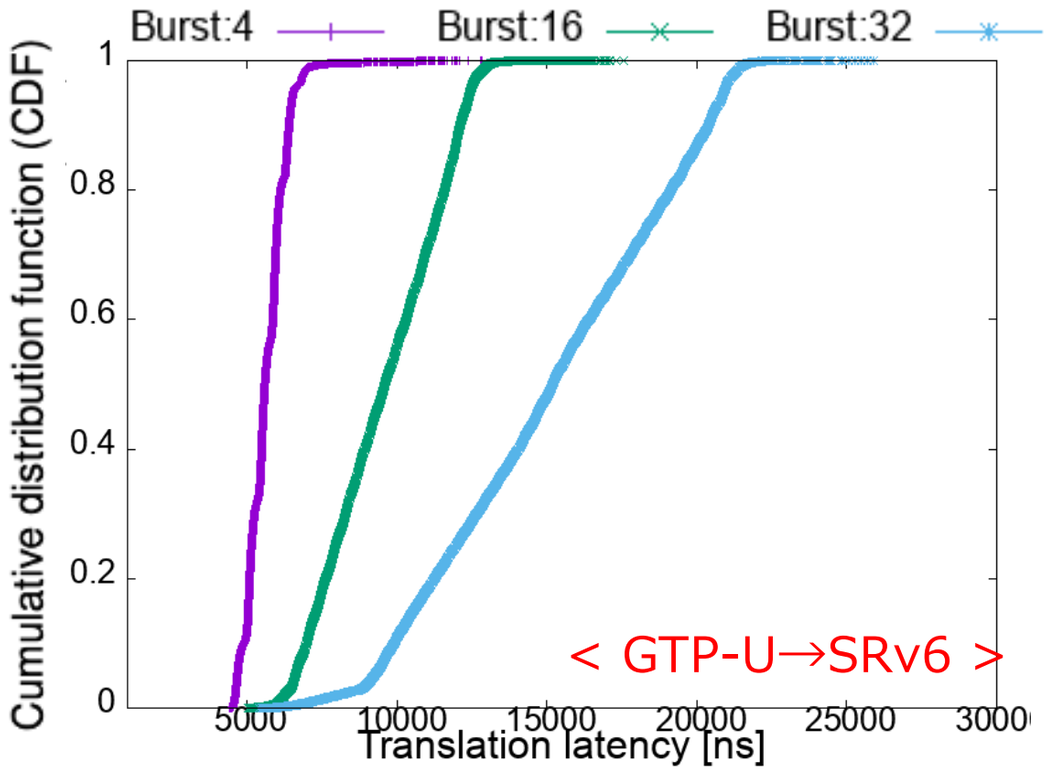
	Packet header	Header	Address
Outer headers	GTP-U over IPv4 (Outer)	SIP	172.20.0.2 (fixed)
		DIP	172.20.0.1 (fixed)
	SRv6 over IPv6 (Outer)	SIP	c1::ac14:2:0:0 (fixed)
		DIP	d4:0:ac14:1::c800:0 (fixed)
	GTP-U (Outer)	TEID	10 (fixed)
Inner headers	IPv4 (Inner)	SIP	1.0.0.1 - 100.0.0.254 (incremental)
		DIP	101.0.0.1 - 200.0.0.254 (incremental)
	TCP (Inner)	SPort	1 - 65535 (random)
		DPort	1 - 65535 (random)

Why the ripple points are occurred?

CDF of translation latency (long packet)

- As the ***tx_burst*** size is changed, the latency is proportionately changed
  - The ***tx_burst*** size is the major cause of ripple points

	tx_burst	Min [us]	50%tile [us]	Mean [us]	Max [us]
GTP-U→SRv6	**4**	**4**	**6**	**6**	**13**
	16	5	10	10	18
	32	5	15	15	26
SRv6→GTP-U	**4**	**4**	**5**	**5**	**11**
	16	4	8	8	16
	32	4	14	14	26



Why the ripple points are occurred?

CDF of translation latency (long packet)

- As the ***tx_burst*** size is changed, the latency is proportionately changed
  - The ***tx_burst*** size is the major cause of ripple points

	tx_burst	Min [us]	50%tile [us]	Mean [us]	Max [us]
GTP-U→SRv6	**4**	**4**	**6**	**6**	**13**
	16	5	10	10	18
	32	5	15	15	26
SRv6→GTP-U	**4**	**4**	**5**	**5**	**11**
	16	4	8	8	16
	32	4	14	14	26

